

संभव BATCH - CRASH COURSE

Mathematics: Mensuration (Hard Level)

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SSC CGL Tier-II Level Exams | 50 Premium Practice Questions | No Answer Key

Q1. The base of a right prism is a regular hexagon of side 5 cm. If its height is $12\sqrt{3}$ cm, then its volume (in cm^3) is:

- (A) 450
- (B) 900
- (C) 1350
- (D) 1800

SSC CGL Tier-II 2020

Q2. A solid metallic sphere of radius 8 cm is melted and drawn into a uniform wire of uniform cross-section of radius 2 mm. The length of the wire (in m) is:

- (A) 170.6
- (B) 160.4
- (C) 175.2
- (D) 165.8

SSC CGL Tier-II 2021

Q3. A right circular cone of height 20 cm and base radius 15 cm is cut parallel to its base at a height of 10 cm from the base. What is the volume of the frustum so formed (in cm^3)?

- (A) $4375\pi/3$
- (B) $4125\pi/3$
- (C) $3375\pi/2$
- (D) $4000\pi/3$

SSC CGL Tier-II 2019

Q4. A field is in the shape of a trapezium whose parallel sides are 200 m and 400 m long, while each of the other two sides is 260 m long. What is the area of the field (in m^2)?

- (A) 60000
- (B) 72000
- (C) 84000
- (D) 96000

SSC CPO Tier-II 2022

Q5. The area of the base of a right pyramid is 400 cm^2 and its height is 15 cm. If the base of the pyramid is a square, then the total surface area of the pyramid is (in cm^2):

- (A) 1000
- (B) 1040
- (C) 1200
- (D) 1400

SSC CGL Tier-II 2020

Q6. A cylindrical tube open at both ends is made of metal. The internal diameter of the tube is 11.2 cm and its length is 21 cm. The metal everywhere is 0.4 cm thick. The volume of the metal is (in cm^3):

- (A) 316
- (B) 306.24
- (C) 280.4
- (D) 291.8

SSC CGL Tier-II 2018

Q7. If the radius of the base of a right circular cylinder is increased by 20% and the height is decreased by 30%, then what is the percentage increase/decrease in the volume?

- (A) Increase of 0.8%
- (B) Decrease of 2%
- (C) Decrease of 0.8%
- (D) Increase of 2%

SSC CHSL Tier-II 2022

Q8. A hemispherical bowl of internal diameter 36 cm contains a liquid. This liquid is to be filled into cylindrical bottles of radius 3 cm and height 12 cm. How many such bottles are required to empty the bowl?

- (A) 24
- (B) 36
- (C) 54
- (D) 72

SSC CGL Tier-II 2021

Q9. Three solid metallic spheres of radii 1 cm, x cm, and 8 cm are melted and recast into a single solid sphere of radius 18 cm. The surface area of the sphere with radius x cm is (in cm^2):

- (A) 324π
- (B) 400π
- (C) 1156π
- (D) 100π

SSC CGL Tier-II 2019

Q10. A right triangular pyramid is cut by a plane parallel to its base at the mid-point of its height. The volume of the upper part is what fraction of the volume of the original pyramid?

- (A) $1/4$
- (B) $1/8$
- (C) $1/2$
- (D) $1/16$

SSC CPO 2021

Q11. The sum of the radii of two circles is 286 cm and the area between the concentric circles is 50336 cm^2 . What are the radii of the two circles (in cm)? (Take $\pi = 22/7$)

- (A) 171 and 115
- (B) 160 and 126
- (C) 140 and 146
- (D) 150 and 136

SSC CGL Tier-II 2022

Q12. The perimeter of a semi-circular path is 144 m. Find the area of the semi-circular path. (Take $\pi = 22/7$)

- (A) 1232 m^2
- (B) 1284 m^2
- (C) 1008 m^2
- (D) 1400 m^2

SSC CHSL 2021

Q13. In a triangle ABC, AB = 13 cm, BC = 14 cm, and CA = 15 cm. A circle is inscribed in it. What is the area of the incircle? (Take $\pi = 22/7$)

- (A) $154/3 \text{ cm}^2$
- (B) $160/3 \text{ cm}^2$
- (C) 154 cm^2
- (D) $616/9 \text{ cm}^2$

SSC CGL Tier-II 2020

Q14. The sides of a quadrilateral ABCD are 6 cm, 8 cm, 12 cm, and 14 cm respectively. If the sum of opposite angles is 180° , find the approximate area of the quadrilateral.

- (A) 100 cm^2
- (B) 90 cm^2
- (C) 95 cm^2
- (D) 85 cm^2

SSC CGL 2023

Q15. The base of a right prism is a triangle whose sides are 8 cm, 15 cm, and 17 cm. Its volume is 480 cm^3 . What is its lateral surface area?

- (A) 320 cm^2
- (B) 400 cm^2
- (C) 360 cm^2
- (D) 280 cm^2

SSC CPO Tier-II 2022

Q16. The total surface area of a solid hemisphere is 1039.5 cm^2 . The volume of the hemisphere is (in cm^3): (Take $\pi = 22/7$)

- (A) 2425.5
- (B) 2530.5
- (C) 2145.5
- (D) 2280.5

SSC CGL Tier-II 2021

Q17. A regular hexagon is inscribed in a circle of radius 10 cm. Find the area of the circle that is outside the hexagon. (Take $\pi = 3.14$, $\sqrt{3} = 1.73$)

- (A) 54.5 cm^2
- (B) 62.5 cm^2
- (C) 48.5 cm^2
- (D) 50.5 cm^2

SSC MTS 2022

Q18. The internal and external radii of a hollow spherical shell are 3 cm and 5 cm respectively. It is melted and recast into a solid cylinder of diameter 14 cm. The height of the cylinder is:

- (A) $8/3 \text{ cm}$
- (B) $4/3 \text{ cm}$
- (C) $2/3 \text{ cm}$
- (D) $5/3 \text{ cm}$

SSC CGL Tier-II 2019

Q19. The length, breadth and height of a cuboidal box are in the ratio 7 : 5 : 3 and its whole surface area is 27832 cm^2 . Its volume is:

- (A) 208010 cm^3
- (B) 288120 cm^3
- (C) 315000 cm^3
- (D) 258040 cm^3

SSC Selection Post 2021

Q20. A well of diameter 3 m is dug 14 m deep. The earth taken out of it has been spread evenly all around it in the shape of a circular ring of width 4 m to form an embankment. Find the height of the embankment.

- (A) $9/8 \text{ m}$
- (B) $11/8 \text{ m}$
- (C) $7/8 \text{ m}$
- (D) $5/8 \text{ m}$

SSC CGL Tier-II 2020

Q21. The volume of a right circular cone is 924 cm^3 . If its height is 18 cm, then the area of its base (in cm^2) is:

- (A) 154
- (B) 132
- (C) 176
- (D) 198

SSC CHSL Tier-II 2022

Q22. A solid metallic cuboid of dimensions $18 \text{ cm} \times 36 \text{ cm} \times 72 \text{ cm}$ is melted and recast into 8 identical solid cubes. What is the ratio of the total surface area of the cuboid to the sum of the total surface areas of all 8 cubes?

- (A) 7 : 12
- (B) 7 : 8
- (C) 4 : 7
- (D) 7 : 16

SSC CGL Tier-II 2021

Q23. The radius of a cylindrical vessel is 7 cm and its height is 10 cm. If the vessel is full of water and a solid metallic sphere of radius 3 cm is dropped into it, what is the volume of water displaced?

- (A) $36\pi \text{ cm}^3$
- (B) $27\pi \text{ cm}^3$
- (C) $45\pi \text{ cm}^3$
- (D) $54\pi \text{ cm}^3$

SSC CPO 2020

Q24. A rectangular sheet of metal 36 cm long and 14 cm broad is rolled along its length to form a cylinder. The volume of the cylinder (in cm^3) is:

- (A) 1445.45
- (B) 1440
- (C) 1444.6
- (D) 1443.2

SSC CGL 2022

Q25. The area of a regular octagon with side 6 cm is:

- (A) $72(1 + \sqrt{2}) \text{ cm}^2$
- (B) $36(1 + \sqrt{2}) \text{ cm}^2$
- (C) $72(1 + \sqrt{3}) \text{ cm}^2$
- (D) $36(1 + \sqrt{3}) \text{ cm}^2$

SSC MTS 2021

Q26. Water flows at the rate of 10 m/min through a cylindrical pipe 5 mm in diameter. How long would it take to fill a conical vessel whose diameter at the base is 40 cm and depth is 24 cm?

- (A) 45 mins
- (B) 51.2 mins
- (C) 48.5 mins
- (D) 55 mins

SSC CGL Tier-II 2018

Q27. A tent is in the form of a right circular cylinder surmounted by a cone. The diameter of cylinder is 24 m. The height of the cylindrical portion is 11 m, while the vertex of the cone is 16 m above the ground. The area of canvas required for the tent is:

- (A) 1320 m^2
- (B) 1155 m^2
- (C) 1240 m^2
- (D) 1350 m^2

SSC Selection Post Phase X

Q28. The perimeter of the base of a right square pyramid is 40 cm. If the height of the pyramid is 12 cm, what is its lateral surface area?

- (A) 240 cm^2
- (B) 260 cm^2
- (C) 300 cm^2
- (D) 320 cm^2

SSC CGL Tier-II 2020

Q29. What is the volume of a tetrahedron whose side is $6\sqrt{2}$ cm?

- (A) 144 cm^3
- (B) 72 cm^3
- (C) 108 cm^3
- (D) 216 cm^3

SSC CHSL 2021

Q30. If a cone is cut into two parts by a horizontal plane passing through the midpoint of its axis, the ratio of the volumes of the upper part and the frustum is:

- (A) 1 : 2
- (B) 1 : 7
- (C) 1 : 8
- (D) 1 : 6

SSC CPO Tier-II 2021

Q31. A metallic solid right circular cone of height 9 cm and base radius 7 cm is melted into a solid sphere. What is the radius of the sphere?

- (A) 4.5 cm
- (B) 5 cm
- (C) 5.25 cm
- (D) 6 cm

SSC CGL 2023

Q32. The radius of a sphere is reduced by 40%. By what percent will its volume decrease?

- (A) 78.4%
- (B) 60%
- (C) 64%
- (D) 84.6%

SSC MTS 2022

Q33. A solid cube is cut into 27 identical cubes. What is the ratio of the total surface area of the original cube to the sum of the total surface areas of all the smaller cubes?

- (A) 1 : 3
- (B) 1 : 9
- (C) 1 : 27
- (D) 1 : 8

SSC CGL Tier-II 2021

Q34. The diagonals of a rhombus are 24 cm and 10 cm. Find the perimeter of the rhombus.

- (A) 48 cm
- (B) 52 cm
- (C) 56 cm
- (D) 60 cm

SSC CPO 2022

Q35. The area of the largest circle that can be drawn inside a square of side 28 cm is:

- (A) 616 cm^2
- (B) 580 cm^2
- (C) 784 cm^2
- (D) 420 cm^2

SSC CHSL 2020

Q36. A room is 12 m long, 9 m wide, and 8 m high. Find the length of the longest pole that can be placed in it.

- (A) 15 m
- (B) 17 m
- (C) 19 m
- (D) 21 m

SSC CGL 2019

Q37. The curved surface area of a cylinder is 594 cm^2 and its volume is 1336.5 cm^3 . What is the height of the cylinder?

- (A) 10.5 cm
- (B) 12 cm
- (C) 14 cm
- (D) 15 cm

SSC Selection Post Phase XI

Q38. A wire is bent to form a square of area 169 cm^2 . If the same wire is bent to form a circle, what is its area? (Take $\pi = 22/7$)

- (A) 215 cm^2
- (B) 215.5 cm^2
- (C) 215.1 cm^2
- (D) 215.09 cm^2

SSC CGL Tier-II 2022

Q39. If the ratio of the volumes of two spheres is 64 : 27, then the ratio of their surface areas is:

- (A) 16 : 9
- (B) 4 : 3
- (C) 8 : 3
- (D) 9 : 16

SSC MTS 2021

Q40. The area of a triangle whose sides are 11 cm, 60 cm, and 61 cm is:

- (A) 330 cm²
- (B) 350 cm²
- (C) 420 cm²
- (D) 610 cm²

SSC CPO 2020

Q41. A metallic solid right circular cylinder of radius 3 cm and height 5 cm is melted and recast into a solid cone of base radius 1 cm. The height of the cone is:

- (A) 135 cm
- (B) 45 cm
- (C) 15 cm
- (D) 90 cm

SSC CGL Tier-II 2018

Q42. The difference between the circumference and radius of a circle is 37 cm. The area of the circle is: (Take $\pi = 22/7$)

- (A) 154 cm²
- (B) 144 cm²
- (C) 111 cm²
- (D) 256 cm²

SSC CHSL 2022

Q43. The lengths of two parallel sides of a trapezium are 18 m and 24 m. If its area is 210 m², the distance between the parallel sides is:

- (A) 8 m
- (B) 10 m
- (C) 12 m
- (D) 14 m

SSC CGL 2021

Q44. What is the area of a sector of a circle of radius 14 cm and central angle 45° ?

- (A) 77 cm^2
- (B) 154 cm^2
- (C) 88 cm^2
- (D) 112 cm^2

SSC Selection Post 2021

Q45. A swimming pool is 20 m long, 15 m wide, and 3 m deep. Find the cost of repairing the floor and walls at the rate of Rs. 25 per square meter.

- (A) Rs. 12750
- (B) Rs. 13500
- (C) Rs. 14200
- (D) Rs. 15600

SSC CGL Tier-II 2019

Q46. A sphere is placed inside a right circular cylinder so as to touch the top, base, and lateral surface of the cylinder. If the radius of the sphere is r , the volume of the cylinder is:

- (A) $\frac{4}{3} \pi r^3$
- (B) $2\pi r^3$
- (C) πr^3
- (D) $\frac{8}{3} \pi r^3$

SSC CPO Tier-II 2022

Q47. If the altitude of an equilateral triangle is $12\sqrt{3} \text{ cm}$, then its area is:

- (A) $144\sqrt{3} \text{ cm}^2$
- (B) $72\sqrt{3} \text{ cm}^2$
- (C) $216\sqrt{3} \text{ cm}^2$
- (D) $288\sqrt{3} \text{ cm}^2$

SSC MTS 2020

Q48. The base of a pyramid is an equilateral triangle of side 10 cm. If its height is $40\sqrt{3} \text{ cm}$, then its volume is:

- (A) 1000 cm^3
- (B) 1200 cm^3
- (C) 800 cm^3
- (D) 900 cm^3

SSC CGL 2022

Q49. A rectangular piece of paper 22 cm long and 10 cm wide is folded without overlapping to make a cylinder of height 10 cm. Find the volume of the cylinder. (Take $\pi = 22/7$)

- (A) 385 cm^3
- (B) 350 cm^3
- (C) 410 cm^3
- (D) 395 cm^3

SSC CHSL 2021

Q50. The inner circumference of a circular track is 440 m. The track is 14 m wide everywhere. Find the cost of leveling the track at Rs. 5 per m^2 .

- (A) Rs. 33880
- (B) Rs. 34000
- (C) Rs. 35200
- (D) Rs. 36480

SSC CGL Tier-II 2023